Where Does Machine Advice Emerge? An Auditable, Observational Trace Across OLMo-2's Open Training Pipeline

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Abstract

When asked for personal advice, instruction-tuned models tend to produce a recognizable bundle: an empathy expression, emotional validation, professional referral, and structuring language. Is this one "advice template," and where in training does its targeting change? We trace these behaviors across OLMo-2's open post-training pipeline and exact Stage-1 pretraining corpus, each model's exact SFT mixture, the DPO preference mixes, the RLVR sets, and the public Base→SFT→DPO→RLVR checkpoints — using 140 prompts in five matched conditions, five sampled generations each, and prompt-clustered permutation tests. Three findings. **(1) At the preregistered detector threshold, the bundle decomposes.** Empathy expression is higher on emotional non-advice than advice; structuring language is identical on neutral advice; professional referral occurs on 0.80–0.82 of clinical-information prompts; validation alone is elevated against every matched control in both models. A threshold sweep preserves the structuring result but qualifies the empathy and referral contrasts. **(2) Stagewise, every advice-vs-factual gap widens at SFT; DPO acts behavior-specifically.** A stage×condition difference-in-differences (prompt-clustered permutation test) shows all eight Base→SFT changes are positive (unadjusted $p < 0.05$), with six surviving Benjamini–Hochberg correction across the full adjacent-stage grid; DPO significantly further widens the expressive gaps and *collapses* the structuring gap (-0.28–-0.38, $p < 10^{-4}$) but does *not* significantly move the near-ceiling referral gap; no later-stage gap change is detected. **(3) The open data layers corroborate this.** Referral phrasings occur in thousands of exact-SFT assistant turns. Across ∼378k DPO pairs per model, they remain enriched in chosen responses after pair-level testing and word normalization ($q < 10^{-7}$); structuring language is stronger still ($q < 10^{-124}$). Both have near-zero direct lexical evidence in the searched RLVR data; advice answers are also more novel than factual ones against the exact pretraining corpus (verbatim-only, hedged). The evidence is observational, a preliminary detector audit indicates high recall/low precision (rates are upward-biased), and the SFT co-occurrence is phrase-sparse; we report all three plainly and release the tracer, prompts, and artifact.

1 Introduction

Ask a modern instruction-tuned model "I've been feeling really anxious lately and can't sleep, any advice?" and you will, in our sample, reliably receive a structured response: an acknowledgment, validation of the user's emotional state, suggestion to seek professional help, and list-organizing language. Two questions follow. First, is this a single *advice template*, or several behaviors that merely co-occur on advice prompts? Second, *where* in training does each behavior emerge?

Answering either requires public training artifacts and intermediate weights. A final OLMo-2 Instruct checkpoint is the product of four data-bearing stages — pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2024) — and OLMo-2 is, to our knowledge, the only current family for which all post-training data *and* intermediate weight checkpoints are public. We exploit both: we run the same behavioral evaluation on every public stage checkpoint, and we search each stage's exact data — the Stage-1 corpus OLMo-Mix-1124 via infini-gram, each model's exact SFT mixture (`tulu-3-sft-olmo-2-mixture` for the 1124-7B line, `tulu-3-sft-olmo-2-mixture-0225` for the 0425-1B line), the DPO preference mixes, and the RLVR sets.

We are deliberate about what this evidence is: **observational co-occurrence and stage localiza-**

tion, not causal attribution. We measure where behaviors appear across checkpoints and where their phrasings live in the corresponding data; we do not retrain. Our data search is not fully end-to-end: Stage-2 mid-training (Dolmino-Mix-1124) has no public index and is not searched.

Contributions.

1. **An auditable behavior-level tracer on exact open data.** A released lexicon+e5 detector, role-scoped (assistant-only) corpus search, and a 140-prompt battery in five matched conditions (advice, factual, emotional non-advice, neutral advice, domain factual), five seeds per prompt, with prompt-clustered permutation inference.

2. **A controlled decomposition of the "advice template."** At the primary threshold, empathy expression tracks emotional content, structuring language tracks the how-to form, professional referral tracks advice/health, and validation alone is elevated against every matched control in both models; the threshold sweep identifies which distinctions are detector-sensitive (Section 4.1).

3. **A stagewise difference-in-differences with significance.** A prompt-clustered permutation test on per-prompt trajectories across Base→SFT→DPO→RLVR checkpoints shows every gap widens at SFT (six of eight changes survive multiplicity correction), DPO significantly widens the expressive gaps and *collapses* the structuring gap (generalizing lists to non-advice) but not the near-ceiling referral gap, with no later-stage gap change detected (Section 4.2).

4. **Data-layer corroboration with paired inference.** Referral phrasings cluster in exact-SFT assistant turns (phrase-sparse, leave-one-out reported) and remain enriched in DPO chosen responses under pair-level McNemar tests and word normalization; searched RLVR counts are near zero. A low-precision detector and verbatim-only novelty are stated as first-class limitations (Sections 4.3–4.4).

2 Related Work

Behavioral tests and checkpoint trajectories. Our matched prompt battery follows the behavioral-testing principle that targeted capabilities require controlled tests beyond aggregate accuracy (Ribeiro et al., 2020). Pythia makes dense public checkpoints available for training-dynamics research (Biderman et al., 2023); checkpoint studies have tracked how loss and downstream behavior change during pretraining (Xia et al., 2023). We instead compare named post-training stages and test changes in condition gaps rather than raw behavior rates.

Pretraining attribution and n-gram search. infini-gram (Liu et al., 2024) makes exact-substring counting over web-scale corpora tractable, and OLMoTrace (Liu et al., 2025) traces model outputs back to pretraining text. Both are pretraining-only; we add role-scoped search over the post-training layers, where our stagewise results locate the change.

Open instruction and preference data. Tulu-3 (Lambert et al., 2024) is the recipe behind OLMo-2's post-training; the SFT mixtures, DPO preference mixes, and RLVR sets are public, as is oasst1 (Köpf et al., 2023) (itself an internal subset of the Tulu-3 mixtures). Public data at every stage is what makes the trace auditable.

The superficial alignment hypothesis. LIMA (Zhou et al., 2023) argues alignment largely surfaces a latent style from few demonstrations; Ghosh et al. (2024) interrogate this. Our stagewise gaps give a behavior-level version: each behavior exists at base, and post-training mainly *re-targets* it toward the right conditions.

Causal data attribution. LESS uses optimizer-aware gradient similarity to select influential instruction data (Xia et al., 2024); Xiao and Aranguri (2026) perform probe-based causal attribution on OLMo-2 DPO and mitigate a behavior by editing top-attributed preference pairs. We differ in kind: our trace is observational and cheap, covers all four stages and both data and checkpoints, and *nominates* clusters and stages for exactly such interventions; it does not establish causation.

3 Method

3.1 Behavior detection (lexicon + e5)

A hand-seeded, human-auditable lexicon defines four reported constructs: **empathy expression** (artifact key `empathy_opener`), **validation**, **professional referral** (`disclaimer`), and **structuring language** (`structure`). A fifth category, `crisis_referral`, fails the preliminary audit and is excluded. Exact matching is augmented with **e5-small** paraphrase embeddings (Wang et al., 2024), a category firing when any sentence exceeds similarity $\tau = 0.82$ to a seed exemplar (sweep $0.80/0.82/0.84$). **Detector construction and circularity.** The lexicon, exemplars, and threshold were hand-seeded during an 8-prompt pilot on OLMo-2-1B and frozen *before* the present 140-prompt battery was authored; all three control conditions and 100 of the 140 prompts were written after freezing. The detector is preliminarily audited in Section 4.4 and released in full (Appendix B).

3.2 Prompt battery and generation

Five conditions isolate the axes the "advice template" conflates: **advice** (60 personal/emotional help requests), **factual** (30 knowledge questions), **emotional non-advice** (20 descriptive prompts about emotional topics, no help request), **neutral advice** (20 how-to requests with no emotional stakes), and **domain factual** (10 impersonal clinical-information questions). Each prompt is answered with **five sampled generations** (temperature 0.7, top-p 0.9, 256 new tokens, distinct seeds).

3.3 Statistical unit

The clustering unit is the **prompt**: each prompt contributes its mean emission over its five generations, and condition rates are means over prompts. All between-condition inference uses **prompt-clustered permutation tests** (10,000 permutations of prompt-level means; two-sided; Monte Carlo $p = (b + 1)/(B + 1)$). Benjamini–Hochberg correction is applied separately to the 24 matched-control, 28 adjacent-stage, and 8 DPO pairwise families. We do *not* treat generations as independent Bernoulli trials: Fisher/Wilson statistics computed on "equivalent counts" are reported only as an appendix sensitivity analysis with that caveat (Appendix D).

3.4 Stagewise evaluation and stage-data search

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct; 7B Base→SFT→DPO→Instruct. Base checkpoints have no chat template and use a plain Question/Answer format (a comparison caveat we state). The stagewise grid runs advice+factual; all five conditions run on the final models. For each stage's data we search: the exact Stage-1 corpus (`v4_olmo-mix-1124_llama`; novelty = 1−verbatim coverage), each model's exact SFT mixture in DuckDB with **assistant-only** scoping (866,138 / 939,344 conversations for 1B/7B; oasst1 not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares behavioral phrases against **length-matched topical controls** from the same answers, per-million documents, with a distinct-phrase leave-one-out.

Reproducibility. Every number derives from `summary_v3.json` and released scripts under `experiments/`; prompts, lexicon, seeds, thresholds, and adjusted-p output ship with it.

4 Results

4.1 The advice bundle decomposes at the primary threshold

Table 1 gives mean prompt-level emission per condition on the final models; Table 2 tests advice against each matched control. Against *plain factual* controls every behavior separates sharply in both models (all prompt-clustered $p < 10^{-4}$; referral 0.927 vs 0.073 at 1B, 0.913 vs 0.060 at 7B). But the matched controls show this is **not one advice-specific template**:

- **Empathy expression tracks emotional content at $\tau = .82$.** It fires *more* on emotional non-advice descriptions than on advice (0.79 vs 0.587 at 1B, $p = .005$; 0.76 vs 0.617 at 7B, $p = .098$), and near-zero on neutral advice (0.03–0.07).
- **Structuring language tracks the how-to form.** It is at ceiling on *neutral* advice exactly as on personal advice (diff 0.00, $p = 1.0$ at 1B; -0.03, $p = .57$ at 7B).
- **Professional referral tracks the advice/health domain at $\tau = .82$.** It fires on 0.66–0.72 of neutral-advice and 0.80–0.82 of clinical-information prompts; its advice increment over domain-factual is small and not significant ($+0.127$, $p = .053$ at 1B; $+0.093$, $p = .159$ at 7B).
- **Validation is the only behavior consistently elevated on personal advice relative to ev-**

ery matched control: higher on advice than *every* control in *both* models (all unadjusted $p \leq .02$; all BH-adjusted $q \leq .028$), including emotional non-advice (0.36 vs 0.16, $p = .008$; 0.383 vs 0.13, $p = .003$).

Thus the primary-threshold "template" separates into affect-linked empathy, form-linked structuring, domain-linked referral, and validation consistently elevated on advice. These are not detector-invariant universals: the threshold sweep preserves the structuring result but qualifies the empathy and referral contrasts (Section 4.4). The 1B-vs-7B differences are small and mixed; we make no scaling claim.

4.2 Stagewise: every gap widens at SFT; DPO acts behavior-specifically

Raw advice-side rates rising across stages would be ambiguous — post-training could simply make *all* outputs more cautious and listy. The right quantity is the **advice-minus-factual gap** Δ_s at each stage (Table 4) and its *change* across adjacent stages, a stage×condition difference-in-differences tested with a prompt-clustered permutation test on per-prompt stage trajectories (Table 3; 10k permutations, n=60 advice/30 factual). Three findings.

- **SFT widens every gap.** All eight base→SFT changes are positive (raw $p < 0.05$; referral $+0.33$/$+0.28$, structuring $+0.39$/$+0.23$). BH correction across all 28 adjacent-stage×behavior tests preserves six; validation is marginal ($q = .087/.055$). Every behavior is already nonzero at base, so SFT re-*targets* a pretrained capability rather than creating it.
- **DPO acts behavior-specifically.** It significantly further widens the *expressive* gaps (validation $+0.16$/$+0.15$, $p \leq 0.008$; 1B empathy $+0.15$, $p = 0.005$) but *not* the referral gap ($+0.10$/$+0.08$, $p = 0.067/0.109$), which is already near ceiling on advice after SFT. Its preference data nonetheless contains more referral phrasings in chosen than rejected responses (Table 5), consistent with *maintaining* that near-saturated gap.
- **DPO collapses the structuring gap.** Its gap falls -0.38/-0.28 at DPO ($p < 10^{-4}$) — not because advice-side structuring drops (it is at ceiling) but because DPO raises structuring on *factual* answers from 0.07 to 0.54–0.59. Preference tuning spreads structuring language everywhere while leaving the safety and expressive behaviors condition-targeted.

RLVR and the final merge move no gap significantly (every dpo→rlvr and final-merge transition has $p \geq 0.21$), consistent with RLVR's math-only data (Section 4.3).

4.3 The data layers corroborate the checkpoint story

SFT: a finite, auditable referral cluster (phrase-sparse). Assistant-only search of each model's exact mixture surfaces thousands of referral demonstrations: "healthcare professional" (3075/3106 for 1B/7B), "mental health professional" (2109/2120), "professional help" (1765/1774), "seek professional" (1353/1355). Length-matched recoverability is above topical controls at ≥ 3 words (3-word: $19.5\times$ at 1B, $7.3\times$ at 7B; 4-word: $141.7\times/14.5\times$) but **phrase-sparse**: only 5/6 distinct 3-word types, dominated by the list preamble "here are some." The distinct-phrase leave-one-out is the honest figure: dropping that one phrase lowers the 3-word mean from 3358.1 to **758.8**/M (1B) and 3026.0 to **1065.9**/M (7B) (Appendix E). Any inference must use the 5–6 distinct types as its unit, not the 46/55 occurrences. Role-scoping remains necessary: blob search inflates "talk to a" 1.41–$1.43\times$ via user turns.

DPO: paired and length-normalized enrichment. The exact mixes contain 378,301/378,341 preference pairs. Chosen responses are slightly *shorter* in aggregate (chosen/rejected mean-word ratio 0.990/0.997), ruling out a general length explanation. At pair level, professional-referral language occurs in 2478 vs 2098 chosen-only/rejected-only pairs at 1B and 2602 vs 1925 at 7B (exact McNemar, BH $q = 4.1\times10^{-8}$ / 2.1×10^{-23}); per-million-word enrichment remains $1.12\times/1.25\times$. Structuring language is stronger still (27047 vs 21779 and 28466 vs 19838; $q < 10^{-124}$; word-normalized $1.16\times/1.27\times$). Empathy is weakly enriched only at 1B after correction, and validation is not significant (Table 5). These paired co-occurrences corroborate, but do not attribute, the checkpoint changes.

RLVR: a clean negative. The RLVR datasets (GSM/MATH/IF) contain essentially none of the advice phrasings: every referral, empathy, and validation phrase counts 0–1 across `RLVR-GSM`, `RLVR-MATH`, and the IF-constraints set (only

Behavior	Model	Advice	Factual	Emotional	Neutral adv.	Domain fact.
empathy expr.	1B	0.587	0.033	**0.790**	0.030	0.340
	7B	0.617	0.027	**0.760**	0.070	0.360
validation	1B	**0.360**	0.000	0.160	0.000	0.120
	7B	**0.383**	0.013	0.130	0.030	0.060
referral	1B	0.927	0.073	0.510	0.660	0.800
	7B	0.913	0.060	0.540	0.720	0.820
structuring	1B	1.000	0.587	0.820	**1.000**	0.940
	7B	0.970	0.547	0.830	**1.000**	0.840

Table 1: Mean prompt-level emission per condition (final models, $\tau = 0.82$; each prompt averaged over 5 generations). Bold marks the cells that break the "advice-specific" reading: empathy peaks on emotional *non-advice* prompts, and structuring language is at ceiling on *neutral* advice. Rates are upward-biased (Section 4.4); `crisis_referral` is excluded (failed preliminary audit).

Behavior	M	−Emot.	−Neut.adv.	−Dom.fact.
empathy	1B	$-.203$ (.005)	.557 ($<10^{-4}$)	.247 (.010)
	7B	$-.143$ (.098)	.547 ($<10^{-4}$)	.257 (.031)
validation	1B	.200 (.008)	.360 ($<10^{-4}$)	.240 (.020)
	7B	.253 (.003)	.353 (.0001)	.323 (.004)
referral	1B	.417 ($<10^{-4}$)	.267 ($<10^{-4}$)	.127 (.053)
	7B	.373 ($<10^{-4}$)	.193 (.001)	.093 (.159)
structuring	1B	.180 ($<10^{-4}$)	.000 (1.0)	.060 (.017)
	7B	.140 (.004)	$-.030$ (.574)	.130 (.026)

Table 2: Advice minus each matched control (prompt-clustered permutation p, 10k draws). Benjamini–Hochberg correction across the 24 displayed tests leaves the validation conclusion unchanged (all $q \leq .028$); non-significant and negative cells carry the decomposition (Section 4.1).

Model	Behavior	base→SFT	SFT→DPO
1B	empathy	$+0.197$ ($<10^{-3}$)	$+0.153$ (.005)
1B	validation	$+0.110$ (.040)	$+0.157$ (.007)
1B	referral	$+0.333$ ($<10^{-4}$)	$+0.097$ (.067)
1B	structuring	$+0.393$ ($<10^{-4}$)	-0.377 ($<10^{-4}$)
7B	empathy	$+0.277$ ($<10^{-3}$)	$+0.070$ (.251)
7B	validation	$+0.120$ (.024)	$+0.147$ (.008)
7B	referral	$+0.277$ ($<10^{-4}$)	$+0.083$ (.109)
7B	structuring	$+0.233$ ($<10^{-3}$)	-0.277 ($<10^{-4}$)

Table 3: Change in the advice-minus-factual gap across transitions (plus-one-corrected prompt-clustered permutation p, 10k draws). BH correction across all 28 adjacent-stage×behavior tests preserves six of eight SFT effects (validation: $q = .087/.055$) and all stated DPO effects; later raw $p \geq .21$.

generic markers appear: "first," 881, "here are some" 48 in the IF set). We find no direct lexical evidence that the searched RLVR sets introduce these behaviors. Their stability through the RLVR stage (Table 4) is consistent with retention rather than new introduction at RLVR.

Pretraining: not a verbatim echo (secondary, hedged). Against the exact Stage-1 corpus, advice answers are more novel than factual ones (0.559 vs 0.383 at 1B; 0.525 vs 0.379 at 7B). This is verbatim-only, length-confounded, and whole-answer rather than behavior-span, so we read it only as: advice answers are not verbatim OLMo-Mix copies. The base checkpoints' nonzero behavior rates are the stronger evidence that the *capability* predates post-training.

4.4 Detector audit and threshold sensitivity

A tool-assisted preliminary pass over 80 sentences drawn from advice answers (Table 6) indicates that the detector is **high-recall, low-precision** (macro P 0.33 / R 0.81; referral P 0.19 / R 1.00). The provenance record does not establish completed independent human review, so we treat these numbers as a diagnostic audit, not human validation. Consequently all emission rates in this paper are **likely upward-biased** (not a formal upper bound: recall < 1). The advice-vs-factual *direction* rests on factual answers triggering fewer false positives, which this advice-only sample **suggests but does not measure**; therefore even the direction of the cross-condition gaps remains provisional. `crisis_referral` failed this preliminary audit outright (0 TP / 16 FP) and is excluded everywhere. We release a **stratified 120-sentence validation set spanning all five conditions and a blinded two-human workflow** (40/30/20/20/10 per condition) with separate files, agreement, confidence intervals, and adjudication. The completed human result remains future work; no cross-condition precision is claimed here. The threshold sweep is informative rather than uniformly reassuring (Appendix C): structuring's advice–neutral contrast stays within .07 in all six model-threshold cells and validation exceeds every control in five of six, but the empa-

Model	Stage	Empathy	Validation	Referral	Structuring
1B	base	.207 (.194)	.073 (.073)	.473 (.413)	.730 (.397)
1B	sft	.390 (.390)	.183 (.183)	.753 (.746)	.857 (.790)
1B	dpo	.590 (.543)	.353 (.340)	.917 (.844)	1.000 (.413)
1B	rlvr	.597 (.550)	.373 (.373)	.903 (.816)	1.000 (.400)
1B	instruct	.587 (.554)	.360 (.360)	.927 (.854)	1.000 (.413)
7B	base	.217 (.217)	.070 (.063)	.543 (.490)	.847 (.480)
7B	sft	.493 (.493)	.183 (.183)	.767 (.767)	.787 (.714)
7B	dpo	.597 (.564)	.330 (.330)	.923 (.850)	.977 (.437)
7B	instruct	.617 (.590)	.383 (.370)	.913 (.853)	.970 (.423)

Table 4: Stagewise advice emission with the **advice-minus-factual gap in parentheses** ($\tau = 0.82$). The referral gap widens at SFT ($+0.33/+0.28$, $p < 10^{-4}$; interaction tests in Table 3) and is not significantly changed thereafter; the structuring gap collapses at DPO because factual answers become listy too. Base checkpoints use a plain QA format (no chat template).

M	Behavior	C-only	R-only	word C/R	BH q
1B	empathy	1768	1634	1.04	.036
1B	validation	183	162	1.12	.282
1B	referral	2478	2098	1.12	$< 10^{-7}$
1B	structuring	27047	21779	1.16	$< 10^{-124}$
7B	empathy	1734	1622	1.05	.074
7B	validation	191	156	1.23	.078
7B	referral	2602	1925	1.25	$< 10^{-22}$
7B	structuring	28466	19838	1.27	$< 10^{-300}$

Table 5: DPO pair-level and length-normalized analysis. C/R is the ratio of phrase occurrences per million whitespace-delimited response words; q is Benjamini–Hochberg adjusted across all eight exact McNemar tests.

category	P	R	F1	tp/fp/fn
empathy expr.	0.25	1.00	0.40	1/3/0
validation	1.00	0.33	0.50	1/0/2
referral	0.19	1.00	0.32	3/13/0
structuring	0.23	0.89	0.36	8/27/1
macro P 0.33 / R 0.81 / F1 0.40; exact-match 0.53; n=80				

Table 6: Preliminary detector P/R/F1 against a tool-assisted diagnostic pass over 80 advice sentences ($\tau = 0.82$). These are not completed human-validation metrics; rates are upward-biased. `crisis_referral`: 0 TP / 16 FP, dropped.

thy emotional–advice contrast reverses at .84 and the referral contrast changes magnitude. We therefore treat the decomposition as a primary-threshold result, not a detector-invariant taxonomy.

5 Discussion

From "advice template" to trained targeting. The decomposition and stagewise gaps suggest a simple account: pretraining supplies each behavior as a *capability*; SFT strengthens targeting (every gap widens; six of eight changes survive correction); DPO then acts selectively — significantly widening the expressive gaps and collapsing the structuring gap (spreading organizers to factual answers) while leaving the already-saturated referral gap unmoved, even as referral phrasings show strong paired, word-normalized chosen-response enrichment; no RLVR-stage gap change is detected. Every step is observational, but each is now separately auditable and significance-tested — and the DPO chosen/rejected differential and RLVR zero-counts are corroborations that pure output-level analysis could not have provided.

Safety implications. The professional-referral behavior — the bundle's safety-relevant component — localizes to a finite, inspectable SFT demonstration cluster and measurable DPO chosen-response enrichment. That makes it auditable, and also fragile: mixture or preference-annotation changes could silently weaken it. The causal leave-cluster-out test our trace nominates (remove the referral cluster, re-SFT, compare against random-cluster removal) is the direct check.

Methodological lessons. (1) Occurrence-weighted provenance ratios are fragile: report distinct phrase types and a leave-one-out. (2) Advice-side rates across stages are ambiguous; report stage×condition *gaps*. (3) Whole-conversation search miscredits user turns; scope to assistant turns. (4) Cluster inference at the prompt, not the sample.

6 Limitations

- **Observational throughout.** Stage localization and data co-occurrence, including the DPO chosen/rejected differential, establish no causation; the nominated leave-cluster-out re-SFT is future work.

- **Detector and threshold.** A tool-assisted preliminary pass suggests high recall and low precision (referral P 0.19), but is not completed human validation; the factual false-positive rate is unmeasured (a provenance-safe, stratified two-human workflow is released). Rates are upward-biased. Absolute levels and the empathy/referral decomposition are threshold-sensitive; structuring and validation are more stable.

- **Coverage gaps.** Stage-2 mid-training (Dolmino-Mix-1124) is unsearched (no public index); matched controls exist only at the final stage; base checkpoints use a different (plain QA) prompt format.

- **Phrase sparsity.** The SFT co-occurrence rests on 5–6 distinct 3-word types dominated by one list marker; use the leave-one-out figures.

- **Statistical scope.** We apply Benjamini–Hochberg correction across the full 28-test adjacent-stage$\times$behavior grid and, separately, across the 24 matched-control tests. Six of eight Base$\rightarrow$SFT changes survive; the two validation changes are marginal. Five seeds bound but do not eliminate sampling variance; prompt counts are modest; one model family, two sizes, one decoding configuration; no scaling claim.

- **Novelty** is verbatim-only, length-confounded, and whole-answer.

7 Conclusion

Machine advice in OLMo-2 is not one template but a bundle of separately-driven behaviors — affect-linked empathy, form-linked structuring, domain-linked referral, and validation consistently elevated against every matched control — whose *targeting* is strengthened at SFT (six of eight changes survive multiplicity correction), then selectively adjusted by DPO (which widens the expressive gaps, collapses the structuring gap by spreading organizers to factual answers, and maintains the near-ceiling referral gap while paired, word-normalized DPO data enrich referral phrasings in chosen responses), with no subsequent gap change detected at RLVR and no direct lexical evidence of introduction in the searched RLVR data. These results are auditable across public post-training stages and selected pretraining layers; we release the tracer, battery, and machine-readable results. The nominated next steps are causal and detector-side: leave-cluster-out re-SFT against random-cluster controls (cf. Xiao and Aranguri, 2026), completion of the stratified two-annotator validation, and a self-hosted Stage-2 index.

Ethics Statement

Data and human subjects. This work uses only publicly released artifacts (OLMo-Mix-1124 via the public infini-gram API; the OLMo-2 Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets; open-weight OLMo-2 models and their public intermediate checkpoints). We collected no new data from people. All 140 prompts are short, author-written strings describing hypothetical situations with no personal information. The study involves no human subjects and needed no ethics-board review.

Sensitive domain. Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns safety-relevant moves such as professional referrals. Given the detector's low precision we report emission rates as upward-biased and drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

Dual use. Making the provenance of a safety behavior inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and release an analysis tool, not any model or dataset that could cause direct harm.

Use of AI Assistants

AI coding and writing assistants were used to implement the analysis tooling, to draft and edit the manuscript, and in the preliminary 80-sentence detector audit. That audit is explicitly not reported as completed human validation. The author designed the study, chose the methodology and metrics, and made the scientific decisions, including dropping `crisis_referral` and choosing the statistical unit. All reported numbers were computed by released code from the released artifact and checked against it.

References

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usvsn Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR.

Sreyan Ghosh, Chandra Kiran Reddy Evuru, Sonal Kumar, S. Ramaneswaran, Deepali Aneja, Zeyu Jin, Ramani Duraiswami, and Dinesh Manocha. 2024. A closer look at the limitations of instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, Shahul ES, Sameer Suri, David Glushkov, Arnav Dantuluri, Andrew Maguire, Christoph Schuhmann, Huu Nguyen, and Alexander Mattick. 2023. OpenAssistant conversations – democratizing large language model alignment. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*.

Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V. Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, Yuling Gu, Saumya Malik, Victoria Graf, Jena D. Hwang, Jiangjiang Yang, Ronan Le Bras, Oyvind Tafjord, Chris Wilhelm, Luca Soldaini, and 4 others. 2024. Tulu 3: Pushing frontiers in open language model post-training. *Preprint*, arXiv:2411.15124.

Jiacheng Liu, Taylor Blanton, Yanai Elazar, Sewon Min, Yen-Sung Chen, Arnavi Chheda-Kothary, Huy Tran, Byron Bischoff, Eric Marsh, Michael Schmitz, Cassidy Trier, Aaron Sarnat, Jenna James, Jon Borchardt, Bailey Kuehl, Evie Yu-Yen Cheng, Karen Farley, Taira Anderson, David Albright, and 11 others. 2025. OLMoTrace: Tracing language model outputs back to trillions of training tokens. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations*, pages 178–188, Vienna, Austria. Association for Computational Linguistics.

Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024. Infini-gram: Scaling unbounded n-gram language models to a trillion tokens. In *First Conference on Language Modeling (COLM)*.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. Association for Computational Linguistics.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, and 23 others. 2024. 2 OLMo 2 Furious. *Preprint*, arXiv:2501.00656. A shorter version appeared at COLM 2025.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual e5 text embeddings: A technical report. *Preprint*, arXiv:2402.05672.

Mengzhou Xia, Mikel Artetxe, Chunting Zhou, Xi Victoria Lin, Ramakanth Pasunuru, Danqi Chen, Luke Zettlemoyer, and Veselin Stoyanov. 2023. Training trajectories of language models across scales. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13711–13738. Association for Computational Linguistics.

Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. 2024. LESS: Selecting influential data for targeted instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning*.

Frank Xiao and Santiago Aranguri. 2026. Probe-based data attribution: Discovering and mitigating undesirable behaviors in LLM post-training. *Preprint*, arXiv:2602.11079. ArXiv preprint, v3.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. 2023. LIMA: Less is more for alignment. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.

A Prompts and decoding configuration

All answers use nucleus sampling (temperature 0.7, top-p 0.9, `max_new_tokens=256`, `do_sample=True`), the chat template where available (plain `Question:/Answer:` for base checkpoints), and five seeds per prompt. The battery: 60 advice, 30 factual, 20 emotional non-advice, 20 neutral advice, 10 domain factual; the complete lists ship with the artifact.

Examples.

- Advice: `I've been feeling really anxious lately and can't sleep. Any advice?`
- Factual: `How does photosynthesis work?`
- Emotional non-advice: `Describe the common symptoms of loneliness.`

- Neutral advice: `How should I organize my bookshelf?`
- Domain factual: `What are common clinical treatments for insomnia?`

B Behavior lexicon and detector threshold

A category fires for an answer if *either* (i) any lexicon phrase occurs as a case-insensitive substring, *or* (ii) any sentence has cosine similarity $\geq \tau$ to a seed exemplar (`intfloat/multilingual-e5-small`, `query:`/`passage:` prefixes), $\tau = 0.82$ primary.

Lexicon (substring match).

- **empathy expression** (`empathy_opener`): `i'm sorry to hear`, `i am sorry to hear`, `i'm really sorry`, `that sounds really`, `that sounds very`, `i can understand`, `i can imagine`, `it's understandable`, `that must be`, `i'm here for you`, `i hear you`, `it sounds like you`
- **validation**: `your feelings are valid`, `it's okay to feel`, `it's normal to feel`, `it's completely normal`, `what you're feeling`, `there's nothing wrong with`, `it's natural to`
- **professional referral** (`disclaimer`): `consult a`, `talk to a`, `speak with a`, `see a doctor`, `seek medical`, `i am not a doctor`, `i'm not a doctor`, `medical professional`, `healthcare professional`, `mental health professional`, `seek professional`, `professional help`
- **crisis_referral** (dropped after the preliminary audit): `crisis`, `hotline`, `helpline`, `988`, `call 911`, `emergency services`, `suicide`
- **structuring language** (`structure`): `here are some`, `here are a few`, `consider the following`, `steps you can take`, `a few things you can`, `first,`, `secondly,`

e5 seed exemplars. empathy: "I'm so sorry you're going through this."; "That sounds incredibly hard." — validation: "Your feelings are completely valid."; "It's totally normal to feel this way." — referral: "You should consult a qualified professional."; "Please talk to your doctor about this." — structuring: "Here are a few steps you can take."; "Consider the following suggestions."

C Threshold sweep

Table 7 reports the exact contrasts underlying the decomposition at $\tau = 0.80/0.82/0.84$. Structuring is stable: advice and neutral-advice remain within .07 in every cell. Validation exceeds its strongest control in five of six cells (the exception is 1B at .80, $-.02$). Empathy's emotional-minus-advice contrast reverses at .84, while referral's advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was frozen before the 140-prompt battery, but its qualitative decomposition should not be generalized across detector thresholds.

M	Contrast	.80	.82	.84
1B	empathy: emot.−adv.	.013	.203	−.127
7B		.030	.143	−.280
1B	validation: adv.−max ctrl.	−.020	.200	.083
7B		.047	.253	.170
1B	referral: adv.−domain	−.003	.127	.163
7B		.017	.093	.280
1B	structuring: adv.−neutral	.000	.000	.067
7B		.000	−.030	−.010

Table 7: Threshold robustness of the four decomposition contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on advice and neutral advice.

D Equivalent-count sensitivity analysis

For readers who want familiar binomial statistics, this table binarizes each prompt's five-generation mean into an equivalent count ($k = \mathrm{round}(\bar{p} \cdot n)$) and applies Wilson intervals, Fisher exact tests, and Haldane–Anscombe odds ratios. **Caveat: these equivalent counts are not independent Bernoulli trials** (each aggregates five correlated generations), so this is a sensitivity analysis only; the prompt-clustered permutation tests in the main text are the primary inference.

M	Behavior	Advice	Factual	OR	Fisher p
1B	empathy	35/60	1/30	27.4	$1.4{\times}10^{-7}$
1B	validation	22/60	0/30	35.6	$4.2{\times}10^{-5}$
1B	referral	56/60	2/30	143.1	$8.9{\times}10^{-17}$
1B	structuring	60/60	18/30	81.8	$3.2{\times}10^{-7}$
7B	empathy	37/60	1/30	31.4	$2.2{\times}10^{-8}$
7B	validation	23/60	0/30	38.2	$1.9{\times}10^{-5}$
7B	referral	55/60	2/30	115.0	$5.7{\times}10^{-16}$
7B	structuring	58/60	16/30	20.6	$1.2{\times}10^{-6}$

Table 8: Equivalent-count sensitivity analysis (advice vs factual, $\tau = 0.82$). Directionally identical to the permutation results; not valid as primary inference (correlated generations within prompts).

E Per-phrase recoverability and leave-one-out

Per-phrase assistant-only counts against each model's exact mixture (`-0225`, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl` = blob/assistant.

Model	Phrase	Asst.	Blob	Infl.
1B	first,	44422	52676	1.19
1B	here are some	11914	13276	1.11
1B	here are a few	5421	6056	1.12
1B	healthcare professional	3075	3216	1.05
1B	mental health professional	2109	2186	1.04
1B	professional help	1765	1825	1.03
1B	seek professional	1353	1377	1.02
1B	it sounds like you	1130	1164	1.03
1B	talk to a	290	410	1.41
1B	it's natural to	152	157	1.03
1B	it's completely normal	78	103	1.32
7B	here are some	12049	13432	1.11
7B	here are a few	5511	6156	1.12
7B	healthcare professional	3106	3309	1.07
7B	i'm really sorry	2369	2378	1.00
7B	mental health professional	2120	2214	1.04
7B	professional help	1774	1838	1.04
7B	seek professional	1355	1379	1.02
7B	it sounds like you	1139	1184	1.04
7B	consult a	927	1031	1.11
7B	talk to a	286	408	1.43
7B	it's completely normal	78	103	1.32

Table 9: Per-phrase SFT-assistant recoverability (representative rows). The 5 (1B) / 6 (7B) distinct 3-word types drive Section 4.3; "here are some" dominates the 3-word mean.

Length-matched buckets (occurrence-weighted). Behavioral vs topical per-million: 1B — 2-word 1839.9 vs 1708.2 (1.1×, n_{occ}=11), 3-word 12333.4 vs 633.8 (19.5×, 46), 4-word 1676.2 vs 11.8 (141.7×, 28); 7B — 2-word 1784.5 vs 996.8 (1.8×, 20), 3-word 9354.4 vs 1280.4 (7.3×, 55), 4-word 1396.6 vs 96.2 (14.5×, 24). **Distinct-phrase leave-one-out (3-word):** dropping "here are some" moves the mean from 3358.1 to 758.8/M at 1B (5 types) and from 3026.0 to 1065.9/M at 7B (6 types).

F Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b+1)/(B+1)$; $p < 10^{-4}$ denotes zero raw exceedances. Appendix D reports Wilson 95% intervals, two-sided Fisher tests, and Haldane–Anscombe odds ratios on equivalent counts. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean behavioral per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs and phrase occurrences per million whitespace-delimited response words. Released scripts derive all statistics from the artifact.